Math 133: Quiz 1

Friday, September 11, 2026 25 minutes 10 points

Name: _____

Show your work for calculations. Give a brief explanation when requested. A simple scientific calculator is allowed, but is not needed.

1. (2 points) The table gives all the input-output pairs for g.

x	-2	0	1	3
$g(x)$	5	2	5	4

   (a) State the domain and range of g.

   (b) Does having $g(-2) = g(1) = 5$ prevent g from being a function? Explain briefly.

2. (2 points) Find the domain of each function. You may use words, inequalities, or interval notation.

   (a) $h(x) = \dfrac{1}{x-4}$ (1 point)

   (b) $k(x) = \sqrt{x+2}$ (1 point)

3. (2 points) A shop's total cost of making q mugs is $C(q) = 30 + 2q$ dollars.

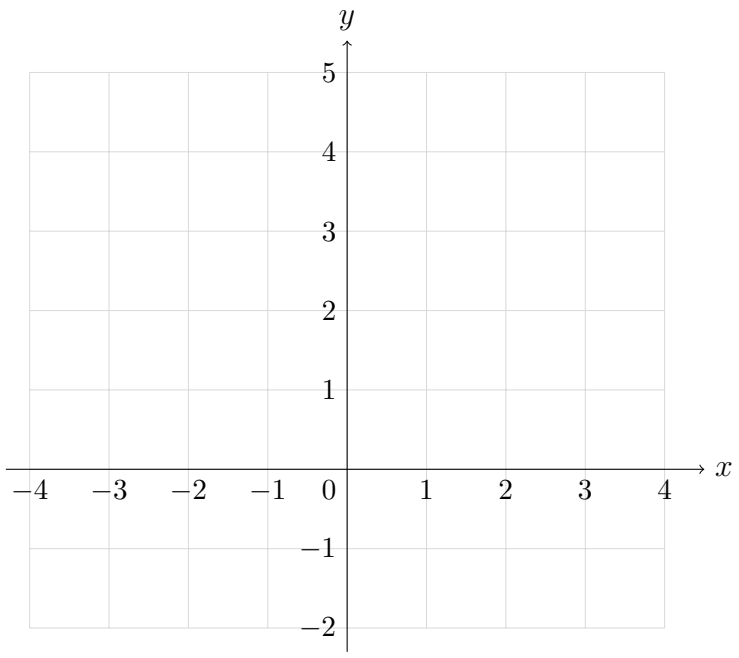
   (a) Calculate $C(10)$ and explain what your answer means in this situation.

   (b) What do the numbers 30 and 2 represent in this model? Include units.

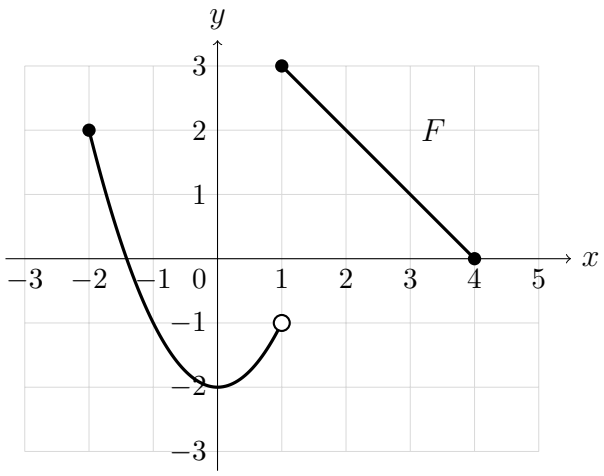
4. (2 points) Graph the piecewise linear function.

$$L(x) = \begin{cases} x + 3, & -3 \le x < -1, \\ 3 - 0.5x, & -1 \le x \le 3. \end{cases}$$

Draw both pieces on the grid. Use open or filled circles to show which endpoints are excluded or included.



5. (2 points) The entire graph of F is shown below. It consists of part of a parabola and a line segment. Open circles are excluded; filled circles are included.



(a) Fill in the table, paying attention to the jump at $x = 1$.

x	-1	0	1	3
$F(x)$				

(b) State the domain and range of F.

(c) Give the open intervals on which F is increasing and those on which it is decreasing.

Increasing: _____

Decreasing: _____